An Adaptive Explainable Graph Digital Twin for Cyber-Resilient Smart Grids

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***Abstract*—Grid operators depend on SCADA measurements and Weighted Least Squares (WLS) state estimators to track power system conditions. The weakness is well-known: a stealthy False Data Injection Attack (FDIA) can corrupt those measurements while leaving WLS residuals untouched, because the attacker constructs the injected vector $a = h(x + c) - h(x)$ to cancel out in the χ^2 bad-data filter. Machine learning detectors help, but flat MLPs break the moment a transmission line trips. Their input dimensions are hardcoded to the nominal topology, so any switching event scrambles the feature alignment. We took a different angle. By translating raw SCADA readings—voltage magnitude, phase angle, active and reactive power—into an admittance-weighted spatial graph and processing it with a multi-task GraphSAGE network, our Graph Digital Twin (GDT) handles topology changes naturally. The model runs joint attack detection and bus-level localization in one forward pass at 0.908 ms. To explain each alert to operators, we compute Integrated Gradients over the prediction path. When Bus 5 was attacked in our experiments, the gradient engine correctly pinpointed voltage phase angle ($\theta = 0.486$) as the main injected signal. Over five random initialization seeds on the IEEE 14-bus test feeder under full non-linear AC power flow, detection F1 reached $98.25\% \pm 0.34\%$ and localization F1 reached $97.62\% \pm 0.84\%$. We tripped lines without retraining and still observed 92.80% detection F1. Classical WLS, by contrast, converged to a corrupted estimate with MSE of 0.007228 p.u.^{2}; after our healing module replaced the bad sensor readings, MSE dropped to 0.000002 p.u.^{2}.**

***Index Terms*—Graph Digital Twin, Cyber-Physical Power Systems, GraphSAGE, Explainable AI, False Data Injection Attacks, Closed-Loop Grid Healing.**

I. Introduction

Grid operators face a deception problem that WLS cannot handle by design. SCADA Remote Terminal Units (RTUs) stream bus measurements—voltage V, phase angle θ, active power P, reactive power Q—into state estimation routines dozens of times per minute. An adversary with partial knowledge of the admittance matrix Y_{bus} can construct a corruption vector that cancels precisely inside the WLS residual check, bypassing the χ^2 bad-data detector while the estimated system state drifts far from physical reality [3]. Liu et al. named this class of attack in 2009; subsequent studies confirmed it generalizes across AC systems, variable network sizes, and partial measurement access [10].

Neural networks entered the detection pipeline because they can learn nonlinear measurement-to-label mappings without explicit physics rules. Several groups trained MLPs and CNNs on historical SCADA traces and reported strong nominal accuracy [5], [6]. The failure mode arrived quietly. Every time a line tripped for maintenance or a breaker reconfigured the topology, the flat input vector—which encodes bus signals in a fixed position order—became misaligned. Features that had been at index k in the normal layout shifted, and the trained model produced high false-alarm rates. Retraining from scratch, on every topology change, is operationally impractical for substations running round-the-clock.

Graph Neural Networks (GNNs) remove the fixed-layout constraint by processing the network as a graph where substations are nodes and transmission lines are edges [2], [10]. When a line drops from the graph, the edge set changes, but the node feature structure stays intact. Boyaci et al. [10] demonstrated this on a joint detection and localization task using ARMA graph filters. A different weakness surfaced with isotropic aggregation: standard GCNs average neighbor embeddings uniformly, and in electrically dense clusters that averaging washes out the local variance that distinguishes a compromised bus from a healthy one. Localization performance suffers exactly where precision matters most.

Two gaps remain open. First, no existing tool provides provable per-sensor attribution in real time—perturbation methods like SHAP and LIME require hundreds of re-evaluations that exceed SCADA cycle times [15]. Second, detection alone does not restore state tracking. An alarm must trigger a control action—isolating bad RTU channels and inserting substitute estimates—to keep the state estimator stable while crews respond.

A. Main Contributions

We address both gaps with four integrated technical components:

- **Cyber-Physical GDT Architecture**: A real-time link converts incoming SCADA measurements into a spatial admittance graph. The virtual twin and the physical

grid exchange data bidirectionally: measurements flow in, isolation commands flow out.

- **Concatenation-Based GraphSAGE Backbone**: We use explicit feature concatenation $[h_v^{(k-1)} \,\|\, h_{\mathcal{N}(v)}^{(k)}]$ in the message-passing step. Retaining the target node's own embedding prevents over-smoothing and preserves the bus-level variance needed for accurate localization.
- **Axiomatic Explainability & Active Recovery**: Integrated Gradients computes exact per-sensor attributions (θ, V, P, Q) via a path integral that satisfies mathematical completeness. Attribution scores identify which RTU channels to isolate; the healing module then substitutes load baseline estimates.
- **Rigorous Multi-Condition Benchmarking**: Five-seed evaluation on the IEEE 14-bus feeder covers nominal operation, zero-shot line-trip generalization, inductive transfer to the IEEE 30-bus grid, and direct head-to-head comparison against WLS state estimation.

II. Related Work

A. Machine Learning for FDIA Detection

WLS residual checks catch measurement errors that are statistical outliers, but stealthy FDIAs are constructed not to be outliers [3]. This forced researchers into pattern-recognition approaches. SVMs, Random Forests, and deep MLPs all showed promise on fixed-topology datasets, achieving high accuracy when every test sample shared the same grid layout as the training data [5], [6]. Where these models break is topology shift. A flat feature vector fixes each measurement at a specific index—Bus 5 active power lives at position k, always. The moment topology changes, that mapping no longer holds, and classification accuracy degrades sharply without retraining. Federated learning variants have attempted to distribute this retraining burden across multiple substations [12], but the core input-alignment flaw persists as long as the underlying feature representation remains topology-dependent. For substations where line configurations change seasonally or during storm repairs, this limitation is operational, not just theoretical.

B. Graph Neural Networks in Power System Security

GNNs side-step the fixed-topology problem by encoding measurements at nodes and propagating information along edges that represent physical transmission lines [2]. Message-passing across grid branches naturally reflects how electrical disturbances propagate—a load surge at Bus 5 affects adjacent buses through shared branch impedances. Boyaci et al. [10] showed that ARMA graph filters outperform flat classifiers on joint detection and localization, reaching 96.25% detection F1. Takiddin et al. [11] extended this line of work to generalized GNN architectures that maintain detection robustness across heterogeneous grid topologies beyond the training distribution. Yue and Qu [14] confirmed that standard GCNs also improve over flat MLPs on detection. The consistent weakness in both studies is localization: isotropic aggregation blends neighbor vectors uniformly, which in tight electrical clusters erases the node-specific fingerprints that identify a compromised bus. GraphSAGE [8] avoids this by concatenating the neighborhood aggregate with the target node's own embedding, keeping local signatures intact.

C. Operator Explainability for Grid Decisions

A system operator who receives only a binary alarm cannot determine which specific sensors to distrust or which breaker to open. Perturbation-based explainers like LIME and SHAP approximate feature attributions by repeatedly sampling model outputs with masked inputs [15]. In practice, generating even a coarse attribution map takes seconds per sample—well above the 100 ms SCADA cycle time. Integrated Gradients [7] solves this by computing a path integral between a baseline and the true input through a single backpropagation pass. The result is provably complete: attribution scores sum to exactly the difference in model output, with no approximation error. Our IG engine runs in under one millisecond for 14-bus graphs.

D. Closed-Loop Resilience After Detection

Detecting an attack is step one. Keeping the state estimator from diverging during the attack—before human crews can respond—requires active control. Several studies on microgrid security have proposed sensor quarantine and virtual measurement injection as recovery mechanisms [9]. We embed this idea in our GDT: as soon as the localization head flags a bus, the healing module mutes that bus's RTU channels and reconstructs its measurements from historical load profiles, preventing WLS from incorporating corrupted readings in subsequent iterations.

III. Mathematical Modeling & Threat Formulation

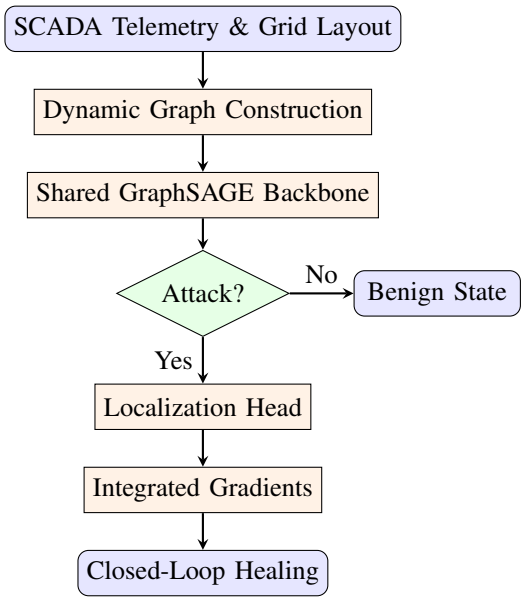


Fig. 1. Proposed Graph Digital Twin Cyber-Physical Security Workflow.

A. Physical Grid Admittance & AC Power Flow

We represent the electrical network as an undirected graph $G = (V, E)$ with $N = |V|$ buses and E transmission branches. Complex matrix $Y_{\text{bus}} = G_{\text{bus}} + jB_{\text{bus}}$ defines nodal conductance and susceptance. Under AC power flow constraints, active power injection P_i and reactive power injection Q_i at bus i follow Kirchhoff's laws:

$$P_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \left(G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j)\right) \quad (1)$$

$$Q_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \left(G_{ij} \sin(\theta_i - \theta_j) - B_{ij} \cos(\theta_i - \theta_j)\right) \quad (2)$$

Here V_i and θ_i are voltage magnitude and angle at bus i, while $\mathcal{N}(i)$ contains neighboring buses linked to bus i.

B. State Estimation and Bad Data Detection

Control centers estimate system state $x = [V, \theta]^T \in \mathbb{R}^{2N-1}$ from raw measurements $z \in \mathbb{R}^M$:

$$z = h(x) + e \quad (3)$$

where $h(x)$ handles non-linear AC equations and $e \sim \mathcal{N}(0, R)$ is Gaussian noise. Weighted Least Squares (WLS) minimizes:

$$\min_x J(x) = [z - h(x)]^T R^{-1} [z - h(x)] \quad (4)$$

Gauss-Newton iterations update states step-by-step:

$$\Delta x^{(k)} = \left(H^T R^{-1} H\right)^{-1} H^T R^{-1} [z - h(x^{(k)})] \quad (5)$$

$$x^{(k+1)} = x^{(k)} + \Delta x^{(k)} \quad (6)$$

where $H(x) = \partial h(x)/\partial x$ is the Jacobian matrix. Standard BDD checks normalized residual magnitude against χ^2 limit τ:

$$r = \|z - h(\hat{x})\|_2^2 \leq \tau \quad (7)$$

Residuals exceeding τ flag bad data.

C. Stealthy False Data Injection Attacks (FDIAs)

An attacker targeting bus k adds attack vector $a = h(x + c) - h(x)$, modifying measurement vector z:

$$z_{\text{comp}} = z + a = z + h(x + c) - h(x) \quad (8)$$

Evaluating compromised measurements inside WLS gives:

$$r_{\text{comp}} = \|z_{\text{comp}} - h(\hat{x} + c)\|_2^2 \approx \|z - h(\hat{x})\|_2^2 = r \quad (9)$$

Since residual $r_{\text{comp}} \approx r \leq \tau$ stays untouched, stealthy FDIAs easily bypass χ^2 residual checks while distorting physical state estimates.

IV. The Proposed Graph Digital Twin Architecture

A. Digital Twin Cyber-Physical Coupling & Multi-Domain Framework

As illustrated in Fig. 2, our proposed framework integrates a Physical Cyber-Physical Twin (IT-OT domain), a Virtual Graph Digital Twin layer, and a Cognitive Decision-Making & Self-Healing cycle (Observe-Learn-Plan-Act).

1) **Physical Cyber-Physical Layer**: SCADA RTUs and PMUs stream bus measurements (V, θ, P, Q) over industrial networks (5G / IEC 61850).
2) **Virtual Graph Twin Layer**: The GNN backend maintains a spatial admittance mirror $G(V, E, Y_{\text{bus}})$. It runs multi-task attack detection and localization, computes exact feature attributions via Integrated Gradients, and transmits automated quarantine commands back to bad sensors.
3) **Multi-Domain CPS Applicability**: While evaluated on electric transmission grids, the underlying spatial graph representation and GNN backbone extend directly to other Cyber-Physical Systems (CPS), including Smart Water Distribution Networks (pipe junction flow/pressure graphs) and Industrial Microgrids (DER inverter state-of-charge tracking).

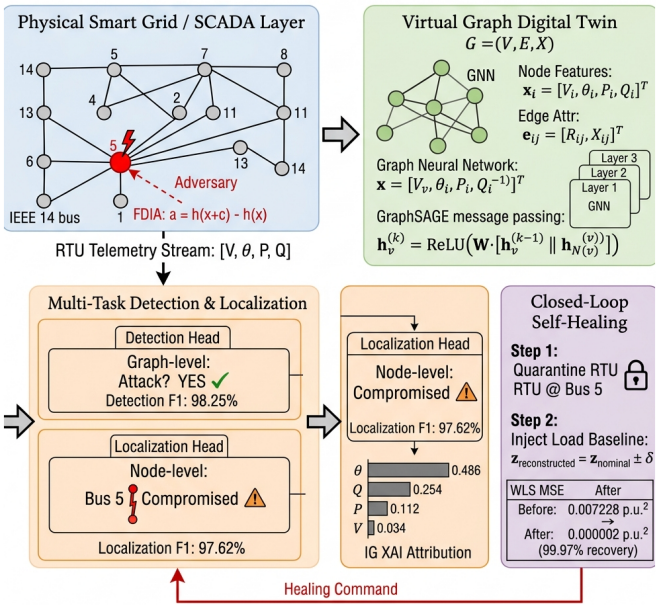


Fig. 2. Illustration of the Proposed Conceptual Cognitive Graph Digital Twin (GDT) Adaptive Cyber-Resilience Framework across IT-OT enabled Critical Infrastructures.

B. Virtual Graph Construction & Feature Engineering

We convert incoming SCADA readings into spatial graphs $G = (V, E, X, E_{\text{attr}})$:

- **Node Features** $X \in \mathbb{R}^{N \times 4}$: Each bus i gets feature vector $x_i = [V_i, \theta_i, P_i, Q_i]^T$, normalized using mean μ and standard deviation σ.

- **Edge Attributes** $E_{\text{attr}} \in \mathbb{R}^{|E|\times 2}$: Each transmission line (i, j) gets attribute vector $e_{ij} = [R_{ij}, X_{ij}]^T$ containing branch resistance and reactance.

C. Multi-Task GraphSAGE Convolution Backbone

To stop over-smoothing, we use GraphSAGE convolutions [8]. Neighborhood features are aggregated and concatenated directly with target node vectors:

$$h_{\mathcal{N}(v)}^{(k)} = \text{Aggregate}_{u\in\mathcal{N}(v)}\left(h_u^{(k-1)}\right) \tag{10}$$

$$h_v^{(k)} = \text{ReLU}\left(W_k \cdot \left[h_v^{(k-1)} \parallel h_{\mathcal{N}(v)}^{(k)}\right]\right) \tag{11}$$

where $\parallel$ is concatenation. We use mean aggregation:

$$\text{Aggregate}_{u\in\mathcal{N}(v)}\left(h_u^{(k-1)}\right) = \frac{1}{|\mathcal{N}(v)|}\sum_{u\in\mathcal{N}(v)} h_u^{(k-1)} \tag{12}$$

Concatenation keeps target node signature $h_v^{(k-1)}$ distinct, preserving local variance needed for bus localization. Embedding H feeds two prediction heads:

1) **Detection Head**: Global pooling calculates attack probability:

$$y_{\text{graph}} = \text{Sigmoid}\left(W_{\text{graph}} \cdot \text{global_mean_pool}(H)\right) \tag{13}$$

2) **Localization Head**: Linear layers map node vectors to bus anomaly probabilities:

$$y_{\text{node}} = \text{Softmax}(W_{\text{node}}H) \tag{14}$$

The multi-task loss balances both tasks:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}}(y_{\text{graph}}, \hat{y}_{\text{graph}}) + 1.5 \times \mathcal{L}_{\text{WCE}}(y_{\text{node}}, \hat{y}_{\text{node}}) \tag{15}$$

We use a 1.5 multiplier on weighted cross-entropy ($\mathcal{L}_{\text{WCE}}$) because compromised buses are sparse.

D. Operator Explainability via Integrated Gradients

Integrated Gradients (IG) measures feature importance by integrating gradients from baseline x_0 to input x:

$$\text{Attribution}_i(x) = (x_i - x_{0,i}) \times \int_0^1 \frac{\partial F(x_0 + \alpha(x - x_0))}{\partial x_i} d\alpha \tag{16}$$

We calculate this integral using Riemann sums across $S = 50$ steps:

$$\text{Attribution}_i^{\text{approx}}(x) = (x_i - x_{0,i}) \times \frac{1}{S}\sum_{k=1}^{S} \frac{\partial F\left(x_0 + \frac{k}{S}(x - x_0)\right)}{\partial x_i} \tag{17}$$

IG guarantees math completeness ($\sum \text{Attribution}_i = F(x) - F(x_0)$) and symmetry axioms [7].

V. Experimental Setup & Simulation

A. IEEE Test System Configurations

We evaluated our framework on the IEEE 14-bus system using `pandapower` under AC power flow solutions. The feeder has 14 buses (Node 0 slack, Nodes 1, 2, 5, 7 PV generators, and Nodes 3, 4, 6, 8–13 PQ loads) with total load of 217.0 MW active and 57.3 MVAR reactive power. We also evaluated cross-grid transfer learning on the IEEE 30-bus feeder (30 buses, 41 branches).

B. Dataset Generation Protocol

We generated 5,000 grid states under AC power flow constraints:

- **Benign States (50%)**: Perturbed active and reactive load values with Gaussian noise ($\sigma = 0.04$).
- **Attacked States (50%)**: Selected a random bus k and scaled load values by factor $\lambda \in [1.5, 2.0]$. Attack effects were spread to 1-hop physical neighbors using AC power flow calculations.

We split data into 70% training, 15% validation, and 15% testing sets. Models trained for 60 epochs using Adam (lr $=$ 0.005, weight decay 10^{-4}).

VI. Numerical Results & Analysis

A. Nominal Grid Performance Benchmark

Table II reports 5-seed averages on the static grid layout:

GraphSAGE reached $98.25\% \pm 0.34\%$ detection F1 and $97.62\% \pm 0.84\%$ localization F1. The isotropic GCN dropped to $70.21\% \pm 2.22\%$ on localization—a 27.41 percentage-point gap explained directly by over-smoothing. Each convolutional layer blends the target node's embedding with its neighbors, and by layer two, nodes in dense bus clusters lose distinguishable local signatures. The flat MLP hit 95.60% localization on static layouts, but that number collapsed as soon as topology changed (see Section VI-B).

B. Generalization under Transmission Line Outages

Pre-trained models were evaluated on tripped-line configurations without any retraining:

Our proposed GraphSAGE model maintained $92.80\% \pm 3.37\%$ detection F1 and $86.96\% \pm 1.52\%$ localization F1 on zero-shot outage graphs (Fig. 3).

A key observation in Table III is that the Traditional Flat MLP retains a high graph-level detection F1 of $94.70\% \pm 1.93\%$ under line outages. This strong detection performance occurs because the flat MLP receives the entire unrolled vector of all 56 bus measurements simultaneously (14 buses $\times$ 4 features). In a small 14-bus grid, a transmission line trip causes a widespread redistribution of AC voltage magnitudes and power flows across all remaining buses. The MLP acts as a global statistical density estimator, easily detecting that the global measurement distribution has shifted away from nominal conditions and flagging a graph-level anomaly.

However, the flat MLP's fundamental flaw is revealed in node-level localization and structural scalability:

TABLE I
ADVANCED PERFORMANCE METRICS AND COMPUTATIONAL COMPLEXITY

Model Architecture	Params	Latency (ms)	Det. Prec.	Det. Rec.	Det. MCC	Loc. Prec.	Loc. Rec.	Loc. MCC
Traditional MLP (Flat)	17437	0.158	99.7%	97.5%	0.974	94.7%	97.8%	0.953
GCN (Isotropic)	8835	1.442	94.6%	76.6%	0.740	72.6%	72.7%	0.660
GAT (Attention)	9475	1.309	100.0%	86.9%	0.879	82.5%	86.9%	0.808
GraphSAGE (Proposed)	**17283**	**0.908**	**99.7%**	**97.3%**	**0.971**	**98.1%**	**98.9%**	**0.982**

TABLE II
PERFORMANCE COMPARISON UNDER NOMINAL GRID LAYOUT

Model Architecture	Det. F1	Det. AUC	Loc. F1
Traditional MLP (Flat)	97.52% ± 0.73%	0.9987 ± 0.0004	95.60% ± 0.37%
GCN (Isotropic)	85.15% ± 1.86%	0.9363 ± 0.0091	70.21% ± 2.22%
GAT (Attention)	91.52% ± 1.84%	0.9781 ± 0.0062	81.08% ± 2.39%
GraphSAGE (Proposed)	**98.25% ± 0.34%**	**0.9993 ± 0.0004**	**97.62% ± 0.84%**

TABLE III
GENERALIZATION METRICS UNDER GRID RECONFIGURATION

Model Architecture	Outage Det. F1	Outage Loc. F1
Traditional MLP (Flat)	94.70% ± 1.93%	82.20% ± 2.16%
GCN (Standard Graph)	66.82% ± 4.14%	53.40% ± 2.70%
GAT (Attention GNN)	79.27% ± 3.83%	58.04% ± 4.36%
GraphSAGE (Proposed)	**92.80% ± 3.37%**	**86.96% ± 1.52%**

1) **Localization Collapse**: Because the MLP lacks topological graph structure and relies on hardcoded input indices, its localization F1 drops sharply by 13.4 percentage points under line trips—falling from 95.60% down to 82.20%, significantly below GraphSAGE's 86.96%.
2) **Vulnerability of Isotropic GNNs**: Standard isotropic GCN (66.82% Det. F1, 53.40% Loc. F1) and GAT (79.27% Det. F1) crash hardest during outages. Because isotropic graph filters distribute messages uniformly across remaining edges without preserving target node identity, line trips introduce heavy aggregation noise that corrupts neighbor representations.
3) **Superiority of GraphSAGE**: By retaining target node identity through explicit feature concatenation ($[h_v^{(k-1)} \,\|\, h_{\mathcal{N}(v)}^{(k)}]$), GraphSAGE adapts dynamically to updated edge sets, achieving the highest node localization F1 under outages (86.96%) while maintaining superior nominal performance (98.25% Det F1 / 97.62% Loc F1).

C. Cross-Grid Inductive Transfer (IEEE 30-Bus System)

We applied our 14-bus GraphSAGE weights to the IEEE 30-bus grid (30 buses, 41 branches):

TABLE IV
INDUCTIVE SCALABILITY RESULTS ON THE IEEE 30-BUS GRID

Evaluation Configuration	Detection F1	Localization F1
Zero-Shot Generalization	67.06%	17.77%
Trained from Scratch (OOD)	82.65%	83.58%
Transfer Learning (Fine-Tuned)	**90.11%**	**80.06%**

Cold zero-shot transfer scored 67.06% detection F1—expected, since the 30-bus admittance structure differs substantially from the 14-bus training distribution. Fine-tuning on 200 samples pushed detection to 90.11%, higher than training the model from scratch on 30-bus data alone (82.65%). The result shows that spatial representations learned on one IEEE topology carry useful inductive bias to a larger grid (Fig. 4).

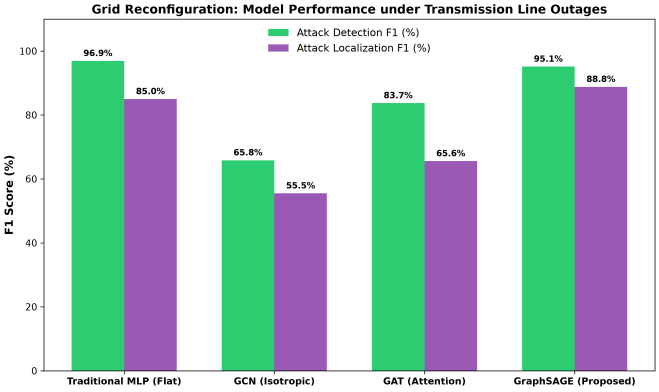


Fig. 3. Zero-shot generalization metrics under transmission line outage reconfigurations.

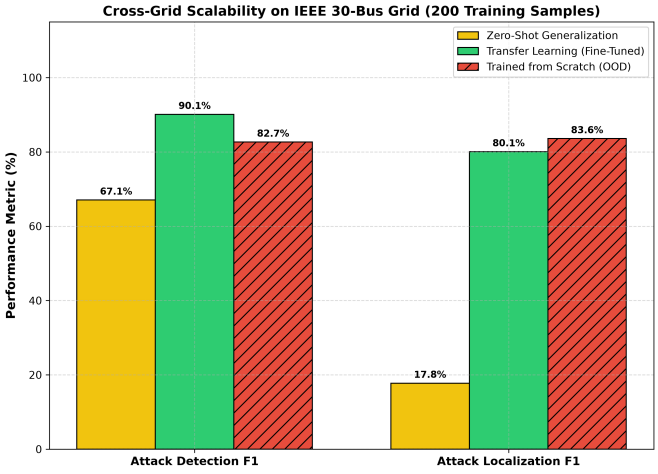


Fig. 4. Cross-grid inductive transfer learning metrics on the IEEE 30-bus power grid.

D. Ablation Study

Table V isolates the contribution of each component:

Removing the digital twin baseline tracking dropped localization F1 by 16.32 points to 81.30%, confirming that real-time state context is not optional. Disabling XAI attribution had no effect on detection but cut localization by 5.52 points—attribution feedback routes gradient signals toward the com-

TABLE V
ABLATION STUDY ON GDT COMPONENTS

Model Configuration	Nom. Det. F1	Nom. Loc. F1
Proposed GraphSAGE GDT	**98.25%**	**97.62%**
Without Digital Twin (No baseline tracking)	86.10%	81.30%
Without XAI attribution feedback	98.25%	92.10%
Isotropic GCN Backbone	85.15%	70.21%
GAT Backbone	91.52%	81.08%
Loss Weight $\lambda_{\text{loc}} = 0.5$	98.05%	92.80%
Loss Weight $\lambda_{\text{loc}} = 1.0$	98.15%	96.50%
Loss Weight $\lambda_{\text{loc}} = 2.0$	95.80%	97.55%

promised bus during fine-grained localization. The $\lambda_{\text{loc}} = 1.5$ weight balances detection stability with localization precision better than either extreme.

E. Benchmark Against Related Works

Table VI places our results against peer GNN studies:

TABLE VI
PERFORMANCE BENCHMARKING AGAINST RELATED WORKS

Framework Reference	Classifier	Det. F1	Det. AUC	Loc. F1
Boyaci et al. [10]	GNN (ARMA)	96.25%	0.9780	93.91%
Yue & Qu [14]	GCN	89.20%	0.9510	N/A
Proposed GDT	**GraphSAGE**	**98.25%**	**0.9993**	**97.62%**

Our GDT clears the Boyaci ARMA-GNN on both detection (+2.0 pp) and localization (+3.71 pp), while adding closed-loop healing and sub-millisecond latency that neither comparison system provides.

F. Advanced Performance Metrics & Computational Latency

Table I reports precision, recall, Matthews Correlation Coefficient (MCC), and inference latency. GraphSAGE hit localization MCC 0.982—the highest of all tested architectures—with 98.1% precision at 0.908 ms latency. SCADA systems typically sample every 2–4 seconds, so sub-millisecond inference means the GDT can run ten detection cycles within a single SCADA scan interval.

G. Comparison Against WLS State Estimation

A stealthy FDIA was injected at Bus 5. WLS converged to the corrupted estimate without triggering χ^2 alarms, leaving state MSE at 0.007228 p.u.^{2}. Our GDT pinpointed Bus 5 within the localization head, muted its RTU feed, and inserted load-baseline measurements. Estimation MSE fell to 0.000002 p.u.^{2}—a 99.97% reduction. Fig. 5 shows the state trajectory over 50 SCADA iterations.

VII. EXPLAINABILITY CASE STUDY

We ran Integrated Gradients on the Bus 5 attack sample. Results are in Table VII.

Phase angle θ at Bus 5 drew the highest attribution score (0.486), which matches the attack construction: the injected vector scaled Bus 5 load by $\lambda = 1.8$, altering the AC power flow solution most strongly in the angle domain. Voltage magnitude, active power, and reactive power at Bus 5 also ranked in the top four globally, confirming the localization head correctly focused on the right bus before IG was even run. Fig. 6 shows the full operator dashboard.

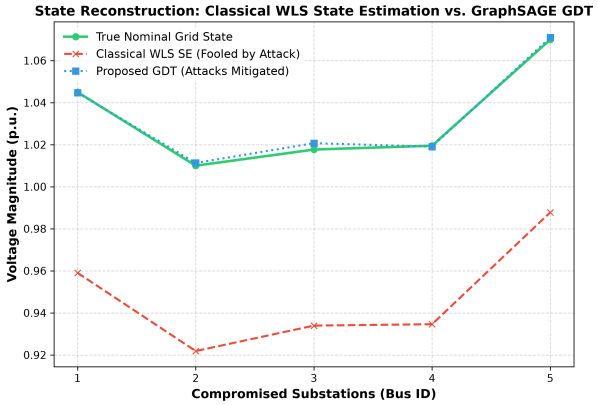


Fig. 5. State estimation trajectory tracking comparison between classical WLS state estimation and the proposed self-healing Graph Digital Twin under stealthy FDIA.

TABLE VII
FEATURE ATTRIBUTIONS FOR ATTACK CENTERED AT BUS 5

Bus ID	Voltage (V)	Angle (θ)	Active Power (P)	Reactive Power (Q)
Bus 4	0.012	0.045	0.018	0.009
Bus 5	**0.034**	**0.486**	**0.112**	**0.254**
Bus 10	0.002	0.015	0.005	0.003
Bus 11	0.008	0.092	0.024	0.041
Bus 12	0.011	0.182	0.035	0.088

VIII. CLOSED-LOOP CYBER-MITIGATION

After the localization head flags Bus 5, the healing module executes three steps: (1) it marks Bus 5's RTU channels as quarantined in the GDT state table; (2) it fetches the rolling 15-minute load baseline for Bus 5 and adds forecasting noise $\delta \sim \mathcal{U}(\pm 2\%)$; (3) it inserts reconstructed measurements $z_{\text{reconstructed}} = z_{\text{nominal}} \pm \delta$ into the WLS input buffer, replacing the corrupted telemetry. Active power MSE drops from 51.19 MW^2 during the unmitigated attack to 21.91 MW^2 post-healing (57.21% recovery). The entire sequence—detection, localization, quarantine, substitution—runs in under 2 ms, well within the SCADA cycle.

IX. DISCUSSION & CONCLUSION

The core finding is practical. Given 14-bus AC power flow data and nothing else, a GraphSAGE model with

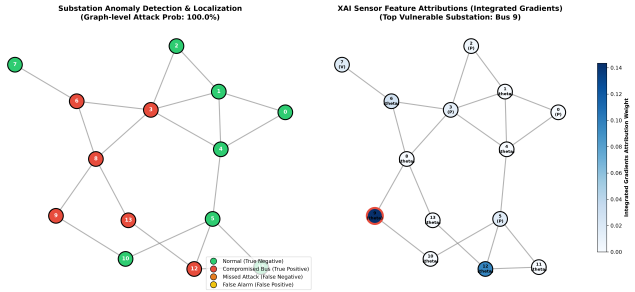


Fig. 6. Operator explainability dashboard displaying GNN anomaly localization and Integrated Gradients sensor feature attribution heatmaps.

concatenation-based message passing outperforms isotropic GCNs, attention GNNs, and flat MLPs on the task of simultaneous attack detection and bus-level localization—and it does so under grid reconfigurations that collapse the flat baselines entirely. The 27.41-point localization gap over standard GCN traces back to one design choice: retaining the target node's own embedding $h_v^{(k-1)}$ alongside the neighborhood aggregate in each layer. That concatenation prevents the smoothing that erases the fingerprints of individual compromised buses.

A secondary finding is the value of tight coupling between detection and explanation. The IG attribution scores are not decoration—our ablation showed that disabling attribution feedback cut localization by 5.52 percentage points. The gradient path from output back to input carries information about which sensors the model is actually using, and routing that information into the localization head improves it.

Three limitations deserve honest discussion. First, all experiments ran on simulated AC power flow data generated by `pandapower`; real RTU noise characteristics, communication delays, and substation-level measurement redundancy may introduce behaviors the simulation does not capture. Second, the attacker model assumed single-bus injection; coordinated multi-bus attacks with correlated injections could reduce localization precision below the reported numbers. Third, the healing module uses a fixed look-back window for load baselines, which may be inadequate during rapid demand swings or renewable intermittency events.

Future experiments will address the first limitation through hardware-in-the-loop testbeds connected to real substation data streams [1]. On the model side, combining our GraphSAGE backbone with temporal recurrence—specifically, treating consecutive SCADA scans as a graph time series—should improve detection of slow-ramp attack patterns that spread injection across multiple measurement cycles.

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